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Master in Business Analytics & Data Science

Research Capstone in Operations Research

Does Spatial Correlation of Carbon Intensity Improve Distributionally Robust Carbon-Aware Data-Center Scheduling?

A Multi-Grid Falsification Study with a Causal Mechanism

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AI Use Declaration. Generative AI tools (Anthropic Claude) were used as a research and engineering assistant under the author's direction: for code implementation support, experiment orchestration, literature triage, and drafting assistance. All research questions, modelling decisions, experimental designs, and interpretations are the author's, developed with the supervisor; all results were produced by version-controlled, pre-registered code reviewed by the author, and all text was reviewed and edited by the author.

Abstract

Carbon-aware schedulers shift data-center compute toward hours and regions where electricity is cleanest, and recent work models carbon intensity as uncertain within a distributionally robust optimization (DRO) framework. A natural next step is to treat carbon intensity as a stochastic *vector* across electrically coupled regions, so that *spatial correlation* can inform the schedule. This thesis asks whether that spatial structure carries any robust scheduling value. Using a Mahalanobis–Wasserstein DRO formulation solved as a second-order cone program, we design a falsification test — the schedule fitted to the joint covariance versus one fitted to a block-diagonal covariance with all cross-region structure destroyed — evaluated by out-of-sample $\mathrm{CVaR}_{0.95}$ of daily emissions under strict pre-registration. Across three primary US/Canada grids spanning the dependence spectrum — the US West (California, Nevada, Phoenix), an Ontario-anchored Eastern Interconnection belt, and an engineered heterogeneous solar/wind/hydro portfolio — corroborated by an Iberia–France panel as a low-correlation anchor, the answer is a replicated, robustness-checked **null**: the spatial gap never exceeds a few tenths of one percent, survives shrinkage and residualization batteries, Benjamini–Hochberg correction, and walk-forward validation. Two diagnostics explain the mechanism. A *mean-ablation* experiment shows the covariance signal is genuine and exploitable but dominated by the mean carbon field; a *tail-dependence* analysis shows the residual dependence is non-elliptical — upper-tail-independent and radially asymmetric — and thus structurally invisible to covariance-based ambiguity sets. A second phase forecloses the obvious objection by replacing the covariance ball with Gaussian and lower-tail (Clayton) copula schedulers that *can* represent the asymmetry: the null survives intact, and a mean-dominance bound grounded in the translation invariance of CVaR explains why no passive dependence model can help. The practical recommendation is that per-region marginal schedulers capture essentially all the value; spatial value, if any, must come from an active transfer channel rather than a richer dependence model.

Contents

1 Introduction

1.1 Context

Data centers consumed an estimated 1–2% of global electricity and their demand is growing rapidly with AI workloads [7]. Unlike most industrial loads, compute is unusually *flexible*: many batch and training workloads can be delayed by hours or moved between sites with little loss of utility. Because the carbon intensity of grid electricity varies strongly by hour and by region — driven by wind, solar, hydro availability, and demand — this flexibility lets a data-center operator act as a grid-aware participant, deferring work from dirty hours to clean ones [11, 14]. Google's carbon-intelligent computing platform demonstrated such temporal shifting at production scale [11], and recent systems work extends it to variable-capacity operation [15].

Scheduling against *tomorrow's* carbon intensity, however, means scheduling under uncertainty. Distributionally robust optimization (DRO) addresses this by optimizing against the worst distribution in an ambiguity ball around the empirical distribution [5], and has been applied to carbon-aware scheduling with carbon intensity as the uncertain quantity [6]. Existing formulations treat each region's carbon intensity essentially in isolation. Yet carbon intensities of electrically interconnected regions co-move: weather fronts traverse neighbouring balancing areas, and shared solar fleets create synchronized clean periods. It is therefore natural to model carbon intensity as a stochastic *vector* with a full spatial covariance — the extension studied here.

1.2 Problem statement and research questions

Modelling spatial dependence is not free: cross-region covariance blocks must be estimated from limited data, and a richer uncertainty model is only worthwhile if it buys measurably better schedules. The literature has assumed, rather than tested, that spatial correlation in the carbon field is worth modelling. This thesis tests that assumption directly.

RQ1. Is the carbon intensity of electrically coupled regions genuinely spatially correlated, beyond shared diurnal cycles?

RQ2. Does feeding that spatial correlation to a covariance-based (Mahalanobis–Wasserstein) DRO scheduler reduce out-of-sample tail emissions ($\text{CVaR}_{0.95}$), relative to an otherwise identical scheduler that ignores cross-region dependence?

RQ3. If not, *why* not — and what dependence structure, if any, would a richer uncertainty model need to capture?

1.3 Contributions

The thesis makes four contributions. (i) *A falsification design*: a shuffled-marginals test that isolates exactly the value of spatial covariance, evaluated out-of-sample under pre-registration discipline (commit-locked configurations, dry-run gates, single test-set read). (ii) *A replicated empirical null across the dependence spectrum*: three primary US/Canada region sets — two interconnected grids where strong correlation survives de-seasonalization (the US West — California, Nevada, Phoenix; and an Ontario–Eastern belt) and one engineered near-uncorrelated solar/wind/hydro portfolio — corroborated by an Iberia–France panel where residual correlation collapses to the diurnal cycle, all yielding spatial gaps below 0.4% of CVaR, robust to estimator choice, multiple-testing correction, and walk-forward validation. (iii) *A causal mechanism*: a mean-ablation experiment showing the covariance signal is real but dominated by the mean field, and a tail-dependence analysis showing the residual dependence is non-elliptical (upper-tail-independent, radially asymmetric), hence invisible to covariance-based ambiguity sets by construction. (iv) *Practice and theory implications*: a per-region marginal scheduler captures essentially all attainable value, and the evidence yields a precise specification mandate for copula-based DRO.

1.4 Thesis structure

Section 2 reviews the literature and locates the gap. Section 3 develops the optimization model, the falsification test, and the evaluation protocol. Section 4 reports the correlation validation and the experimental results across all region sets, including the robustness battery, mean-ablation, tail-dependence diagnostics, and the Phase 2 copula schedulers. Section 5 interprets the findings against prior work and states limitations. Section 6 concludes and outlines future work.

2 Literature Review

2.1 Carbon-aware computing

Carbon-aware scheduling exploits temporal and spatial variation in grid carbon intensity. In production at Google, Radovanović et al. [11] describe a system which computes day-ahead "virtual capacity curves" that throttle flexible compute during high-carbon hours; measured impact on latency-sensitive workloads is negligible while load shifts measurably toward cleaner hours. Wiesner et al. [14] analyse the potential of temporal shifting across regions and workloads, finding the benefit depends strongly on the grid's marginal mix and the workload's flexibility. Wijayawardana and Chien [15] study cloud-VM scheduling on datacenters whose capacity varies with grid conditions — including carbon-coupled capacity, where a constant carbon budget forces capacity to fall as intensity rises — and document the goodput cost of capacity variation for long-running jobs. This systems literature establishes operational levers (deferral, throttling, capacity coupling) that we adopt as constraints, but it treats future carbon intensity as a point forecast; uncertainty is handled implicitly, by re-planning.

2.2 Distributionally robust optimization

DRO optimizes against the worst-case distribution within an ambiguity set, and the Wasserstein ball around the empirical distribution has emerged as the canonical data-driven choice, with tractable finite-dimensional reformulations [5, 9]. For costs linear in the uncertain vector, type-2 Wasserstein DRO with a Mahalanobis ground metric reduces to a mean term plus a covariance-weighted regularizer — an interpretable "mean plus $\varepsilon\sqrt{x^\top \Sigma x}$" objective solvable as a second-order cone program (SOCP) [2]. Hall et al. [6] apply Wasserstein DRO to carbon-aware data-center scheduling with virtual capacity curves, treating each region's carbon intensity as the uncertain quantity and demonstrating robustness gains over sample-average approximation. The risk functional we target, CVaR_β, is standard for tail-focused evaluation [12]. Beyond second moments, recent work robustifies the *dependence structure* itself: Fan et al. [3] place both the marginals and the *copula* within a Wasserstein ball, the natural home for the non-elliptical structure our Phase 2 copula schedulers target.

2.3 Spatial dependence and its modelling

Cross-region dependence enters a Mahalanobis–Wasserstein model only through the off-diagonal blocks of the covariance matrix. Estimating large covariance matrices from short panels is noisy; shrinkage estimators [10] are the standard remedy and serve here as a robustness arm. Beyond second moments, dependence is fully described by the copula [8, 13]; tail-dependence coefficients quantify joint extremes that correlation cannot see, and the Gaussian copula has zero asymptotic tail dependence for any correlation below one [4] — a property we exploit as a diagnostic benchmark. Vine-copula constructions make high-dimensional non-elliptical dependence tractable [8] and are the natural candidate for a dependence-aware extension of DRO ambiguity sets.

2.4 The gap

To our knowledge, no prior work *tests* whether spatial correlation of carbon intensity — however strong in the data — translates into out-of-sample robust scheduling value. The systems literature does not model uncertainty explicitly; the DRO literature models uncertainty per region but has not isolated the marginal contribution of *cross-region* dependence; and neither offers a controlled mechanism for *why* spatial structure should or should not matter. This thesis fills that gap with a falsification test, a replicated multi-grid study, and two mechanism diagnostics (mean-ablation and tail dependence).

3 Methodology

3.1 Research design overview

The design is a controlled computational experiment. We hold the optimization machinery fixed and manipulate exactly one object — the cross-region blocks of the covariance supplied to the scheduler — then measure the consequence on held-out tail emissions. Three primary region sets spanning the dependence spectrum (with two earlier panels corroborating) serve as replications; three nested constraint regimes guard against the result being an artifact of a particular feasible-set geometry; a robustness battery (estimators, multiple-testing correction, walk-forward) guards against statistical artifacts; and two further experiments (mean-ablation; tail dependence) identify the mechanism. Throughout, a pre-registration discipline removes analyst degrees of freedom.

3.2 The scheduling problem

Compute is scheduled across R regions over a $T = 24$-hour day. The decision is $x \in \mathbb{R}_{\geq 0}^{R \times T}$, where $x_{r,t}$ is compute power (MW) in region r at hour t. Carbon intensity $\rho_{r,t}$ (gCO$_2$eq/kWh) is uncertain; realized emissions are linear, $\langle \rho, x \rangle = \sum_{r,t} \rho_{r,t} x_{r,t}$.

3.3 Distributionally robust formulation

We minimize worst-case expected emissions over a type-2 Wasserstein ball $\mathcal{B}_\varepsilon(\hat{\mathbb{P}})$ of radius ε centred on the empirical distribution of the daily carbon field, with Mahalanobis ground metric induced by the empirical covariance $\hat{\Sigma} \in \mathbb{R}^{RT \times RT}$:

$$\min_{x \in \mathcal{X}} \sup_{\mathbb{Q} \in \mathcal{B}_\varepsilon(\hat{\mathbb{P}})} \mathbb{E}_{\mathbb{Q}}\big[\langle \rho, x \rangle\big] = \min_{x \in \mathcal{X}} \langle \bar{\rho}, x \rangle + \varepsilon \sqrt{x^\top \hat{\Sigma}\, x}, \tag{1}$$

by Wasserstein duality for linear costs [2, 5], where $\bar{\rho}$ is the empirical mean field. With the Cholesky factorization $\hat{\Sigma} = LL^\top$ the penalty becomes $\varepsilon \| L^\top \operatorname{vec}(x) \|_2$, so (1) is an SOCP, solved to global optimality (CLARABEL). The radius ε interpolates from the deterministic problem ($\varepsilon = 0$) to increasingly conservative schedules.

Where spatial correlation enters. Partitioning $\hat{\Sigma}$ into $T \times T$ blocks Σ_{rs} by region pair, the quadratic form splits as

$$x^\top \hat{\Sigma} x = \sum_r x_r^\top \Sigma_{rr} x_r + \sum_{r \neq s} x_r^\top \Sigma_{rs} x_s, \tag{2}$$

so the off-diagonal (spatial) blocks are the *only* channel by which cross-region dependence can alter the schedule. The experiment manipulates exactly these blocks.

3.4 Feasible set and constraint regimes

The feasible set $\mathcal{X}$ collects the linear constraints below; introducing an auxiliary flexible-load variable $x^{\mathrm{f}} \in \mathbb{R}^{R\times T}_{\geq 0}$, the scheduler solves (1) over

$$
\begin{aligned}
&0 \leq x_{r,t} \leq \bar{x}_{r,t}, && \forall r,t && \text{(C0: capacity)} && \text{(3a)}\\
&x_{r,t} = \alpha_r W_r\, p_{r,t} + x^{\mathrm{f}}_{r,t}, && \forall r,t && \text{(C2: flex/inflex split)} && \text{(3b)}\\
&\textstyle\sum_t x^{\mathrm{f}}_{r,t} = (1-\alpha_r)\, W_r, && \forall r && \text{(C2: work conservation)} && \text{(3c)}\\
&|x_{r,t} - x_{r,t-1}| \leq \Delta_r, && \forall r,t && \text{(C3: ramp)} && \text{(3d)}\\
&\textstyle\sum_{t\in\mathcal{W}} x^{\mathrm{f}}_{r,t} \geq \gamma\,(1-\alpha_r)\, W_r, && \forall r && \text{(3a: deferral deadline)} && \text{(3e)}\\
&\pi_{r,t}\, x_{r,t} \leq \bar{P}, && \forall r,t && \text{(3b: thermal/PUE)} && \text{(3f)}\\
&\textstyle\sum_{r,t} \bar{\rho}_{r,t}\, x_{r,t} \leq B && && \text{(3d: carbon budget)} && \text{(3g)}
\end{aligned}
$$

where W_r is region r's daily work (utilization fixed at 80%, $W_r = 960$ MWh); $\alpha_r \in \{0.30, 0.50, 0.75\}$ is the inflexible fraction following the fixed intraday shape $p_{r,t}$ ($\sum_t p_{r,t} = 1$); $\Delta_r = 15$ MW/h is the ramp limit; $\mathcal{W} = [0, 7]$ and $\gamma = 0.20$ set the deferral deadline; and $\pi_{r,t} = \mathrm{PUE}_0 + \kappa \max(\mathcal{T}_{r,t} - \mathcal{T}_{\mathrm{set}}, 0)$ is the temperature-coupled power-usage multiplier (data, hence a constant per cell), capped at effective power $\bar{P}$. Every constraint is linear, so $\mathcal{X}$ is polyhedral and the program (1) remains an SOCP.

The capacity ceiling is constant ($\bar{x}_{r,t} = 50$ MW) except under constraint **3c**, the carbon-coupled variable capacity adapted from Wijayawardana and Chien [15], which replaces it with

$$\bar{x}_{r,t} = x_{\min} + (x_{\max} - x_{\min})\, \mathrm{CFE}_{r,t}/100, \tag{4}$$

where $\mathrm{CFE}_{r,t}$ is the training-mean carbon-free-energy share — capacity falls as the grid gets dirtier. The bounds $(x_{\min}, x_{\max}) = (50, 75)$ were calibrated on training data to a loosely binding ("Goldilocks") regime: the constraint binds on 28–37% of region-hours, leaving positive slack and capacity margin, so it shapes the schedule without freezing it.

Three nested regimes report results as a function of operational tightness: **R3** (reference)

Algorithm 1 Shuffled-marginals experiment (one region set)

Input: daily panels $\rho_{1:N}$ (train 2021–2024, test 2025); regimes {R3, R1, R2}; α-levels $\{0.30, 0.50, 0.75\}$; grid $\mathcal{E} = \{0, 0.1, 1, 10, 100, 1000\}$
for each regime, each α, each arm $\Sigma \in \{\text{joint}, \text{shuf}\}$ **do**
  for each $\varepsilon \in \mathcal{E}$ **do** ▷ blocked 5-fold time-series CV
    for each contiguous fold k **do**
      fit $\bar{\rho}, L$ on training folds; solve SOCP (1); record validation $\text{CVaR}_{0.95}$
  $\varepsilon^\star \leftarrow \arg\min_\varepsilon$ mean validation CVaR ▷ per arm, independently
  refit on full training set at $\varepsilon^\star$; evaluate *once* on 2025 ▷ single test read
paired bootstrap (1000 resamples, fixed seed) on test days $\Rightarrow$ 95% CI for the gap (7)
Pre-registration: configuration commit-locked before any test read; `-dry-run` gate runs CV only

imposes C0 + C2 + C3; **R1** (lean) adds the deferral deadline 3a; and **R2** adds the thermal ceiling 3b (climate cases) or the variable-capacity ceiling 3c (interconnection cases). The carbon budget 3d is available but inactive in the reported regimes.

3.5 The falsification test: shuffled marginals

The scheduler is fitted twice, with covariances differing in exactly one respect:

$$\big(\hat{\Sigma}^{\text{shuf}}\big)_{rs} = \begin{cases} \Sigma_{rr}, & r = s, \\ \mathbf{0}, & r \neq s, \end{cases} \tag{5}$$

preserving every region's marginal distribution and within-region autocorrelation while destroying only cross-region dependence. Each fitted schedule x is evaluated by the out-of-sample conditional value-at-risk of daily emissions $e_i = \langle \rho_i, x \rangle$ realized on test day i. We use the Rockafellar–Uryasev form [12]

$$\text{CVaR}_\beta(e) = \min_{\tau \in \mathbb{R}} \left\{ \tau + \tfrac{1}{1-\beta}\, \mathbb{E}\big[(e - \tau)_+\big] \right\}, \qquad \beta = 0.95, \tag{6}$$

estimated as the empirical mean of the worst 5% of test days. The *spatial gap* is the difference between the two arms,

$$\text{gap} = \text{CVaR}_{0.95}\big(e^{\text{shuf}}\big) - \text{CVaR}_{0.95}\big(e^{\text{joint}}\big), \tag{7}$$

positive when the joint covariance helps. The test is a falsification: it is built to detect a spatial benefit if one exists, so a null is informative. Algorithm 1 summarizes the full protocol.

3.6 Data

Carbon intensity is hourly lifecycle intensity from Electricity Maps (academic licence), 2021–2025, 43,824 hourly observations per zone; panels are built on a common local clock per region set, trained on 2021–2024 and tested once on 2025. Three primary US/Canada region sets span the dependence spectrum: **(1)** the *US West* — California, Nevada and Phoenix (CISO, BANC, LDWP, NEVP, AZPS); **(2)** an *Ontario–Eastern belt* (CA-ON, NYISO, MISO, PJM); and **(3)** an *engineered heterogeneous portfolio* — CISO (solar), ERCOT (wind), BPAT (hydro) — deliberately chosen so clean periods do *not* align: the adversarial best case for diversification. Two earlier panels corroborate: an *Iberia–France* European set (ES, PT, FR), where residual correlation collapses to the diurnal cycle, and the California/Nevada subset of the US West before Phoenix was added at the supervisor's request. Temperature for constraint 3b is hourly ERA5 2 m air temperature (Open-Meteo), one load-weighted point per zone. The licensed raw data is never committed; derived summary tables are archived in the repository for traceability.

3.7 Validation and mechanism diagnostics

Correlation validation (RQ1). Raw Pearson correlation conflates co-movement with a shared diurnal cycle. We therefore report correlations of *residuals* after removing each zone's hour-of-day mean (local clock), with overlay plots of standardized intensity for summer and winter weeks.

Robustness battery. Every experiment is re-run with (i) Ledoit–Wolf shrinkage covariance [10]; (ii) covariance estimated from seasonal (monthly-mean) residuals; (iii) covariance from AR(1) residuals. A pre-registered agreement rule counts a spatial effect only if positive and sign-stable across both residual estimators. Bootstrap p-values across all cells receive Benjamini–Hochberg correction at $q = 0.05$ [1]; a walk-forward refit (train 2021–2023, test 2024) checks temporal stability; a log-denser ε grid checks grid sensitivity.

Mean-ablation (RQ3, causal). To test whether the mean field masks the covariance, the mean used for *scheduling* is transformed while evaluation stays on real emissions: LEVEL equalizes regional time-averages (keeping hour-of-day shape); FLAT sets a constant mean, making the mean term constant under fixed demand — a covariance-only world in which any joint-vs-shuf difference is attributable purely to the spatial blocks.

Tail dependence (RQ3, structural). For each region pair we estimate the upper and lower tail-dependence coefficients $\chi_U(q) = \Pr(U > q, V > q)/(1-q)$ and $\chi_L(p) = \Pr(U < p, V < p)/p$ on rank pseudo-observations of residual intensity, comparing χ_U at $q = 0.95$ with the Gaussian-copula value at the matched correlation. Since the Gaussian copula — the dependence family a covariance ball can represent — is asymptotically tail-independent [4], a positive empirical excess would be exploitable tail structure invisible to the DRO; and since elliptical copulas force $\chi_{\mathrm{L}} = \chi_{\mathrm{U}}$, radial asymmetry diagnoses non-elliptical dependence.

3.8 Phase 2: copula dependence models and a mean-dominance bound

The covariance test answers whether *elliptical* dependence helps. Because the tail diagnostic flags *non-elliptical* structure ($\chi_{\mathrm{L}} > \chi_{\mathrm{U}}$), a covariance ball cannot by construction represent it — so a natural objection to the Phase 1 null is "you used the wrong dependence object." Phase 2 forecloses that objection by replacing the covariance ball with copula models that *can* represent the structure, and re-running the same falsification.

Copula-coupled resampling. We generalize the shuffled-marginals test from second moments to the full copula while preserving every region's empirical marginal. Each region r keeps its pool of whole daily profiles $\{\rho_r^d \in \mathbb{R}^T\}$; a scenario is generated by drawing a copula sample $u \in [0,1]^R$ and, for each region, selecting the training day whose daily-mean intensity has rank u_r (low rank = clean day), then taking that day's full profile. The copula thus governs *only* the cross-region coupling, exactly as the shuffled covariance did. Three nested models span the dependence hierarchy: **(i)** the *independence* copula (regions decoupled — the Phase 1 shuffled arm); **(ii)** the *Gaussian* copula at the empirical rank-correlation (elliptical — what the covariance ball can see); and **(iii)** an exchangeable *Clayton* copula, with lower-tail dependence $\lambda_L = 2^{-1/\theta}$ and zero upper-tail dependence [8], fitted by matching mean pairwise Kendall's τ ($\theta = 2\tau/(1-\tau)$) — the object built for the $\chi_{\mathrm{L}} > \chi_{\mathrm{U}}$ finding. Clayton is sampled by Marshall–Olkin frailty; no parametric marginal model is introduced.

Copula CVaR scheduler. For each model we draw $S = 1000$ scenarios $\{\rho^s\}$ and solve the sample-average CVaR program in Rockafellar–Uryasev form [12],

$$\min_{x\in\mathcal{X},\,\tau,\,z\geq 0} \tau + \frac{1}{(1-\beta)S}\sum_{s=1}^{S} z_s \quad \text{s.t.} \quad z_s \geq \langle \rho^s, x\rangle - \tau, \tag{8}$$

over the *same* feasible set $\mathcal{X}$ (3) and regimes as Phase 1, so the only thing that varies across arms is the dependence model. Each schedule is evaluated once on the 2025 panel by out-of-sample $\mathrm{CVaR}_{0.95}$; the reported copula gaps are CVaR(indep) − CVaR(Gaussian), CVaR(indep) − CVaR(Clayton), and the asymmetry increment CVaR(Gaussian) − CVaR(Clayton), each with a paired day-bootstrap CI.

Why the dependence model is second-order: a bound. A structural reason to expect Phase 2 to reproduce the null follows from the translation invariance of CVaR. Writing the carbon field as mean plus zero-mean residual, $\rho = \bar{\rho} + \xi$, daily emissions decompose as $\langle \rho, x \rangle = \langle \bar{\rho}, x \rangle + \langle \xi, x \rangle$, and since $\langle \bar{\rho}, x \rangle$ is deterministic,

$$\mathrm{CVaR}_\beta\big(\langle \rho, x \rangle\big) = \underbrace{\langle \bar{\rho}, x \rangle}_{\text{copula-independent}} + \mathrm{CVaR}_\beta\big(\langle \xi, x \rangle\big). \tag{9}$$

The dependence model affects *only* the residual term, and only through its cross-region aggregation; its total leverage is bounded by the comonotone–diversified spread $\Lambda(x) = \sup_C \mathrm{CVaR}_\beta(\langle \xi, x \rangle) - \inf_C \mathrm{CVaR}_\beta(\langle \xi, x \rangle)$ over copulas C.

Proposition 1 (Mean-dominance) *For any two dependence models, the optimal out-of-sample* $\mathrm{CVaR}_{0.95}$ *differs by at most* $\Lambda(x^\star)$*. The mean-ablation of Section 4.4 measures* Λ *directly: it is the gap recovered when the mean term is removed. Empirically* Λ *is at most a few tenths of one percent of the mean term with the mean present, rising to* $\approx 1.5\%$ *only under flat ablation; hence the mean field pins the schedule and the dependence model — elliptical or not — is immaterial.*

The proposition predicts that even Clayton, the correct object for the empirical asymmetry, cannot recover material value; Section 4.6 confirms it.

3.9 Reproducibility

All code is version-controlled (private repository, continuous-integration test suite, 168 unit tests, including a feasible-set equivalence test pinning the Phase 2 CVaR scheduler to the Phase 1 constraint set); experiment configurations are commit-locked before test-set evaluation; bootstrap and scenario seeds are fixed; and every number reported in Sections 4–5 traces to an archived summary table. Solvers: CLARABEL/ECOS for the SOCP; HiGHS for the CVaR-SAA linear programs and deterministic LP baselines.

4 Results

4.1 The spatial assumption is valid (RQ1)

The premise holds. After removing each zone's hour-of-day mean, cross-region correlations of carbon intensity in the two electrically coupled US/Canada cases are strong and survive de-seasonalization: in the US West, CISO–NEVP 0.78 and CISO–LDWP 0.72 (shared solar resource and weather across WECC); in the Ontario–Eastern belt, CA-ON–NYISO 0.73 and MISO–PJM

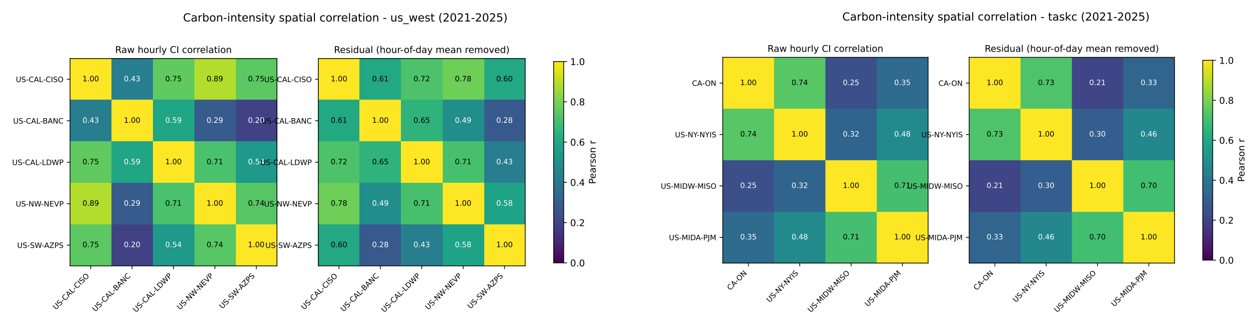

Figure 1: Residual (hour-of-day-removed) cross-region correlation of hourly carbon intensity, 2021–2025. Left: US West (CISO, BANC, LDWP, NEVP, AZPS). Right: Ontario–Eastern belt (CA-ON, NYISO, MISO, PJM). Spatial correlation is strong and genuine in both — the assumption motivating spatial DRO is empirically valid.

0.70 (weather fronts traversing the Eastern Interconnection). The engineered heterogeneous portfolio behaves as designed: ERCOT–BPAT residual correlation is 0.00, with CISO–BPAT at 0.33. Figure 1 shows the residual correlation matrices; overlay plots of standardized intensity (archived with the code) confirm visible co-movement in summer and winter weeks for the coupled cases. Together with the corroborating panels — the California/Nevada subset (0.65–0.78 within-state) and Iberia–France — the cases span the dependence spectrum from ≈ 0 to ≈ 0.8, so whatever the scheduling result, it cannot be attributed to an absence of correlation in the data.

4.2 The spatial gap is null across the spectrum (RQ2)

Table 1 reports the spatial gap (7) for the three pre-registered US/Canada cases: nine cells each (three constraint regimes × three α-levels), CV-selected $\varepsilon^\star$, single test read on 2025. The gap never exceeds a few tenths of one percent of test $\mathrm{CVaR}_{0.95}$ in either direction, and where the bootstrap calls a cell "detectable" the signs flip across adjacent α-levels — the signature of solver-path artifacts rather than exploitable structure. The California/Nevada panel agrees (gaps in $[-0.012, +0.058]\%$, DRO active throughout).

Crucially, this is a null of the *strong* kind. In all three pre-registered cases cross-validation *chooses* a non-trivial ambiguity radius: $\varepsilon^\star = 1$ in every one of the 27 cells for `taskc` and `us_hetero`, and in 25 of 27 for `us_west`. The DRO is therefore genuinely engaged — it is paying for robustness, not collapsing to the nominal problem — and the two arms still coincide. A null in which CV had selected $\varepsilon^\star = 0$ would be uninformative (robustification switched off, so the arms must match by construction); here the robustification is demonstrably on and the spatial covariance still buys nothing. This is the first regime in the project where the DRO is consistently active, which is precisely what makes the null informative rather than vacuous.

The Iberia–France panel is the instructive contrast and the project's historical starting point.

Table 1: Spatial gap (% of test $\mathrm{CVaR}_{0.95}$; positive = joint covariance better) for the three US/Canada cases. R3 = reference regime, R1 = +deadline, R2 = +variable carbon-coupled capacity. "det." = bootstrap 95% CI excludes zero.

		`us_hetero` ($\bar{r} \approx 0$)		`taskc` ($\bar{r} \approx 0.5$)		`us_west` ($\bar{r} \approx 0.5$)	
Regime	α	gap %	det.	gap %	det.	gap %	det.
R3	0.30	−0.233	✓	−0.022	✓	+0.043	
R3	0.50	−0.177	✓	−0.002		+0.032	
R3	0.75	+0.001		+0.034	✓	−0.005	
R1	0.30	−0.233	✓	−0.022	✓	+0.014	
R1	0.50	−0.177	✓	−0.002		+0.028	
R1	0.75	+0.001		+0.034	✓	+0.002	
R2	0.30	−0.211	✓	+0.038	✓	+0.008	
R2	0.50	−0.154	✓	+0.030	✓	+0.005	
R2	0.75	+0.062	✓	+0.009		+0.000	✓

There the residual correlation is near zero, and CV does not even consistently engage the DRO: $\varepsilon^\star$ flickers between 0, 0.1, 1 and 10 across cells, collapsing to 0 in several. This is itself consistent with the thesis — where there is no cross-region structure, the machinery sees nothing worth robustifying — but it makes the Iberian null the *weak* kind. We therefore retain it only as low-correlation, real-grid external validity, and rest the central claim on the US/Canada cases where the DRO is unambiguously on. The engineered `us_hetero` portfolio supplies the strong zero-correlation null ($\varepsilon^\star = 1$ throughout, still no value) that Iberia cannot.

Three readings of Table 1 matter. First, in the adversarial diversification case (`us_hetero`) the joint covariance actually *hurts* at low α (up to −0.23%): with no true cross-structure, the joint estimator fits noise that the block-diagonal arm is spared. Second, in the strongly correlated belt (`taskc`) the detectable cells flip sign between $\alpha = 0.30$ and $\alpha = 0.75$ within the same regime. Third, the US West — strongest correlation of all — produces no detectable cell beyond one of magnitude $+3 \times 10^{-6}$%. Figure 2A plots all 27 cells against each case's mean residual correlation: the cloud stays on zero across the whole correlation range. **Validity of the spatial assumption does not imply value from it.**

4.3 The null survives a pre-registered robustness battery (RQ2)

Table 2 summarizes. Re-estimating the covariance with Ledoit–Wolf shrinkage, seasonal residuals, or AR(1) residuals never produces a spatial effect that passes the pre-registered agreement rule (positive and sign-stable across both residual estimators). The largest battery value anywhere, +0.179% (AR(1), `taskc`, $\alpha = 0.30$), appears *only* under AR(1) and not under the seasonal estimator, so the rule rejects it. Benjamini–Hochberg correction at $q = 0.05$ across all 144 non-

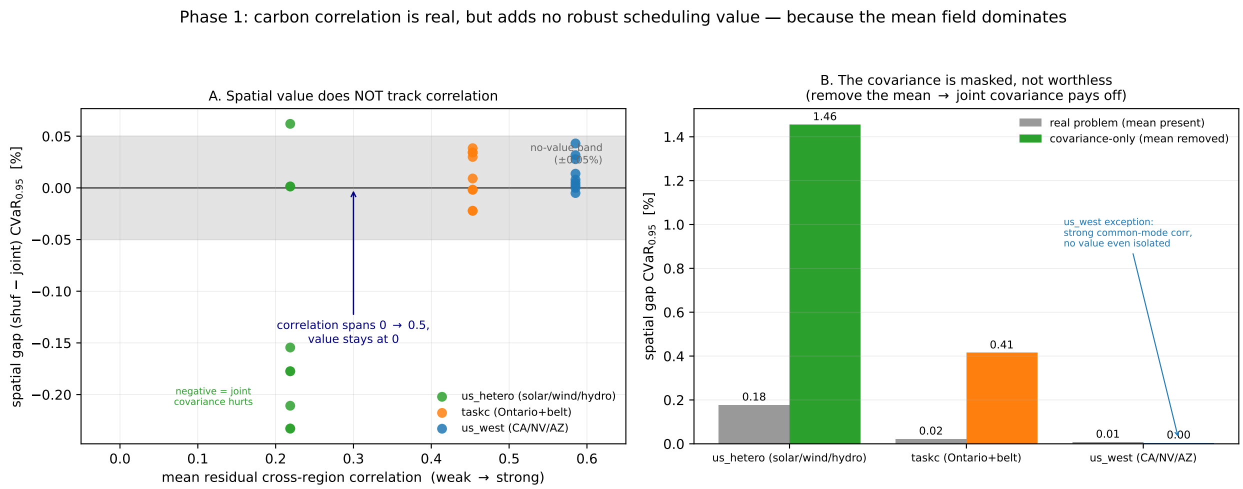

Figure 2: The Phase 1 finding. **A:** spatial gap vs. mean residual cross-region correlation — value does not track correlation anywhere on the spectrum. **B:** mean-ablation — in the covariance-only world (constant scheduling mean) the joint covariance pays off by up to +1.46%, proving the spatial signal is real but *masked* by the mean carbon field, not absent.

Table 2: Robustness battery: maximum |gap| (%) per covariance estimator, and temporal walk-forward (test year 2024). No cell passes the pre-registered agreement rule (positive and sign-stable under both seasonal and AR(1) residualization).

Case	Ledoit–Wolf	Seasonal	AR(1)	Walk-forward 2024
`us_hetero`	0.233	0.355	0.035	0.217
`taskc`	0.037	0.034	0.179	0.036
`us_west`	0.043	0.066	0.017	0.028

ablation cells leaves that same already-rejected cell as the largest surviving positive gap — nothing material survives both filters. A walk-forward refit (train 2021–2023, test 2024) reproduces the null out of time: maximum absolute gap 0.036% (`taskc`), 0.028% (`us_west`), 0.217% (`us_hetero`, again negative). An 11-point log-denser ε grid changes no gap at the reported precision.

4.4 Mean-ablation: the covariance is masked, not worthless (RQ3)

The ablation isolates the cause (Table 3, Figure 2B). Under LEVEL ablation — regional time-averages equalized, hour-of-day shape kept — every gap is unchanged to three decimals: the *between-region* mean differences are not what masks the covariance. Under FLAT ablation — constant scheduling mean, the covariance-only world — the joint covariance suddenly pays: up to +1.46% in `us_hetero` and +0.41% in `taskc`, all BH-significant, and CV no longer collapses both arms onto the same schedule. The spatial signal in the covariance is therefore *real and exploitable*; in the real problem it is simply dominated by the within-day shape of the mean carbon field, which

Table 3: Mean-ablation: spatial gap range (%) across the nine cells per case. LEVEL (equalize regional time-averages) leaves the null untouched; FLAT (constant scheduling mean = covariance-only world) reveals genuine spatial value except where correlation is common-mode (`us_west`).

Case	Real problem	LEVEL	FLAT (covariance-only)
`us_hetero`	$-0.233 \ldots +0.062$	unchanged	$+0.39 \ldots +1.46$
`taskc`	$-0.022 \ldots +0.038$	unchanged	$+0.14 \ldots +0.41$
`us_west`	$-0.005 \ldots +0.043$	unchanged	$-0.04 \ldots +0.00$

Table 4: Residual tail dependence at $q = 0.95$ for selected pairs. "Excess" = empirical χ_U minus the Gaussian-copula value at the matched correlation ρ. Throughout, χ_U shows no exploitable upper-tail excess while $\chi_L > \chi_U$ signals non-elliptical (radially asymmetric) dependence.

Case	Pair	ρ	χ_U emp	χ_U Gauss	Excess	χ_L emp
`us_west`	CISO–LDWP	0.72	0.25	0.41	−0.16	0.48
`us_west`	BANC–LDWP	0.65	0.17	0.35	−0.18	0.40
`us_west`	CISO–NEVP	0.78	0.43	0.47	−0.05	0.39
`taskc`	CA-ON–NYISO	0.73	0.38	0.42	−0.04	0.31
`taskc`	MISO–PJM	0.70	0.43	0.39	+0.04	0.42
`us_hetero`	CISO–BPAT	0.33	0.10	0.16	−0.06	0.26
`us_hetero`	ERCOT–BPAT	0.00	0.04	0.05	−0.02	0.02

both arms share. The instructive exception is `us_west`: even in the covariance-only world its gaps stay at zero (−0.036 to +0.004%), because its strong correlation is *common-mode* — all five zones move together, so destroying cross-dependence changes no scheduling decision. Positive correlation is non-diversifiable risk: there is no hedge for a DRO to find.

4.5 Tail dependence: the structure is non-elliptical (RQ3)

Table 4 reports residual tail-dependence at $q = 0.95$ for representative pairs. Two findings. *First*, the upper tail is empirically at or below the Gaussian benchmark everywhere (excess −0.18 to +0.04): in the dirtiest hours, regions *de-couple* rather than spike together — there is no upper-tail co-movement for a richer ambiguity set to exploit on the risk side. *Second*, the dependence is radially asymmetric, $\chi_L > \chi_U$: regions go *clean* together more strongly than dirty together, clearest in the solar-driven US West (CISO–LDWP $\chi_L = 0.48$ vs. $\chi_U = 0.25$; BANC–LDWP 0.40 vs. 0.17). Elliptical copulas force $\chi_L = \chi_U$, so this is structure a Mahalanobis–Wasserstein ball is geometrically incapable of representing — the covariance is not merely masked (Section 4.4); for the part of the dependence that varies, it is also the *wrong object*. Figure 3 shows the asymmetry for the Ontario–Eastern belt.

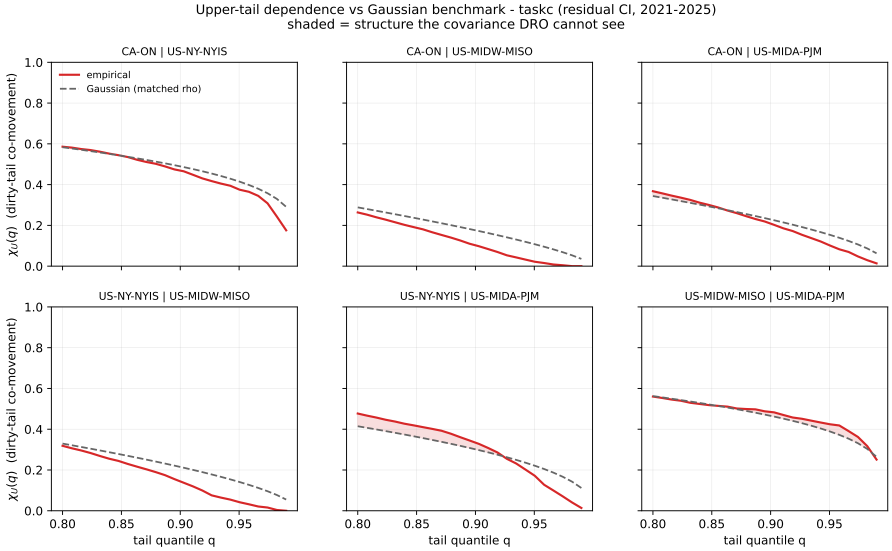

Figure 3: Upper-tail dependence $\chi_U(q)$ for the Ontario–Eastern belt region pairs (solid) against the Gaussian-copula benchmark at the matched correlation (dashed). The empirical curves sit at or below the benchmark as $q \to 1$: in the dirtiest hours the regions decouple, so there is no exploitable upper-tail co-movement for a richer ambiguity set to capture. The complementary radial asymmetry ($\chi_L > \chi_U$) is quantified in Table 4.

Table 5: Phase 2 copula results. Gap = reduction in out-of-sample $\mathrm{CVaR}_{0.95}$ versus the independence baseline (%; positive = the structured copula helps); "Clay−Gauss" is the mean asymmetry increment. No case shows a material, sign-stable copula gain. θ is the fitted Clayton parameter.

Case	$\bar{\tau}$	θ	$\max\lvert\mathrm{gap}_{\mathrm{Gauss}}\rvert$	$\max\lvert\mathrm{gap}_{\mathrm{Clay}}\rvert$	mean (Clay−Gauss)	detectable
`us_west`	0.47	1.77	0.066	0.038	+0.011	none stable
`taskc`	0.31	0.90	0.158	0.083	−0.056	none stable
`us_hetero`	0.20	0.50	0.270	0.127	+0.070	none stable

4.6 Phase 2: the null survives richer dependence models (RQ3)

If the covariance ball failed only because it is elliptical, a copula that encodes the empirical $\chi_L > \chi_U$ asymmetry should recover value. It does not. Figure 4 confirms the Clayton copula fitted to each grid behaves as intended — its samples pile into the joint-clean corner ($\chi_L = 0.70$ vs. $\chi_U = 0.26$ for CISO–BANC), exactly the structure the symmetric Gaussian copula ($\chi_L \approx \chi_U \approx 0.46$) and any covariance ball cannot represent. Yet when each dependence model is used to schedule and evaluated out-of-sample on 2025 (Table 5, Figure 5), no model beats the independence baseline by a material, sign-stable margin: across both arms the largest absolute copula gap anywhere is 0.27% of $\mathrm{CVaR}_{0.95}$ (the Gaussian arm on `us_hetero`; the lower-tail Clayton arm peaks at 0.13%), with signs flipping across α exactly as in Phase 1. The null is now *total* — it holds not just for the covariance ball but for the full copula.

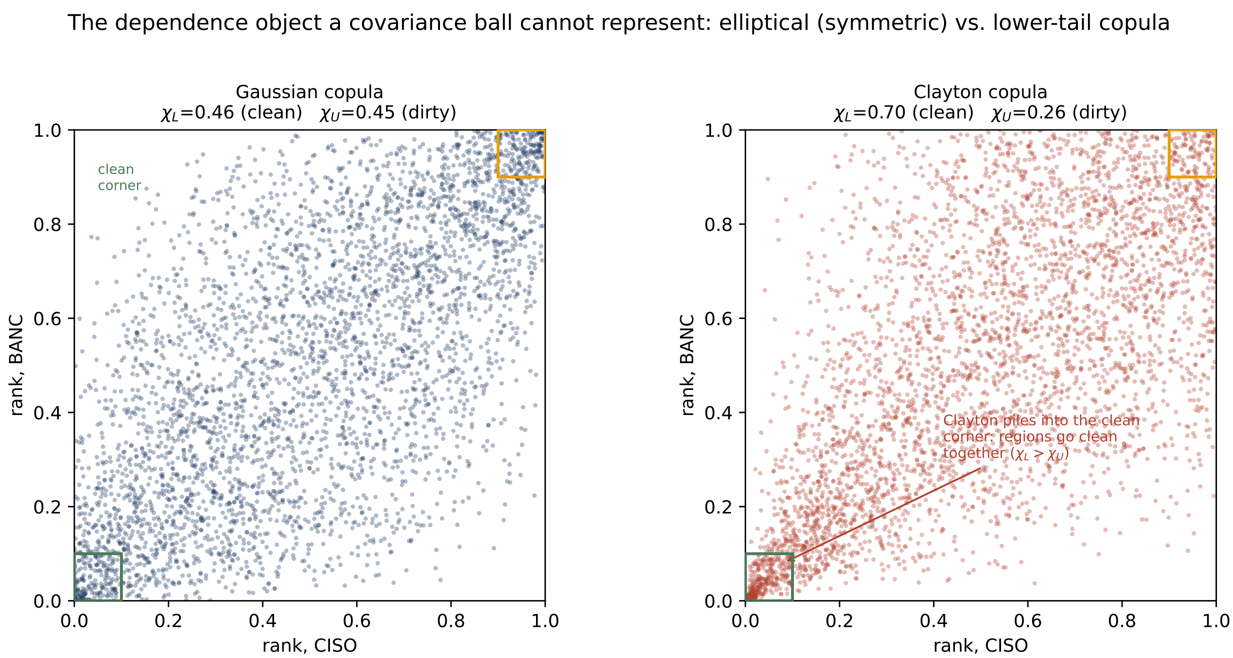

Figure 4: The dependence object a covariance ball cannot represent. Samples from the Gaussian copula (left, symmetric: $\chi_L \approx \chi_U$) and the fitted Clayton copula (right) for a representative US-West pair. Clayton concentrates in the lower-left "clean" corner ($\chi_L > \chi_U$), reproducing the Phase 1 radial asymmetry that elliptical models force to zero.

One positive signal is worth stating precisely: the Clayton−Gaussian increment is consistently positive in the heterogeneous case (`us_hetero`, mean +0.07%) and negligible or negative where correlation is common-mode. The asymmetric copula therefore *does* extract something the elliptical one cannot, in the expected direction — but at a magnitude an order below materiality. This is the empirical face of Proposition 1: the copula's leverage Λ is real but dominated by the mean field. Phase 2 thus converts the Phase 1 null from "covariance is the wrong object" into the stronger "*no* dependence model recovers value here, because the binding constraint is the mean field, not the shape of the dependence."

5 Discussion

5.1 What the null means, and what it does not

The result is a *specification* finding, not a failure of carbon-aware scheduling. Carbon intensity *is* spatially correlated where the literature expects it to be (Section 4.1), and the DRO machinery *is* active (CV selects $\varepsilon^\star = 1$ almost everywhere). What the experiment falsifies is the specific hypothesis that encoding that correlation in a covariance-based ambiguity set improves robust

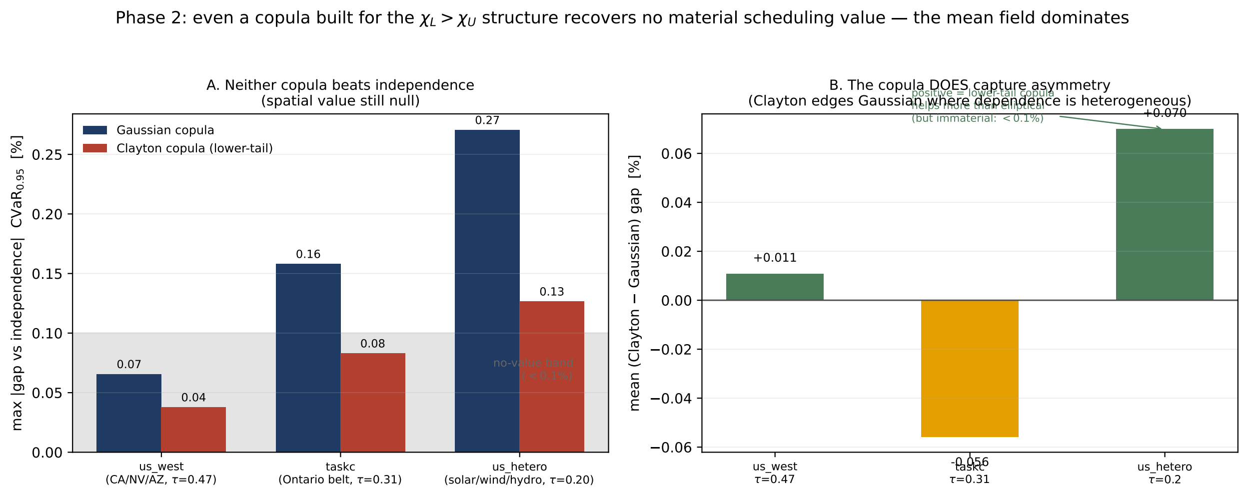

Figure 5: Phase 2 result. **A:** maximum |gap| versus independence for the Gaussian and Clayton arms — neither copula materially beats the decoupled baseline. **B:** the mean Clayton−Gaussian increment by grid; it is positive where dependence is heterogeneous (`us_hetero`, +0.07%), confirming the copula *does* capture the asymmetry — but the effect is an order of magnitude too small to matter, as Proposition 1 predicts.

scheduling. The mean-ablation result (Section 4.4) is what licenses the stronger claim: the covariance signal is genuine and, in a covariance-only world, worth up to 1.46% of tail emissions — it is simply dominated by the diurnal mean carbon field in the problem as actually posed. A reader could otherwise dismiss the null as "the data had no signal"; the ablation shows the signal is present but *subordinate*. This is the central intellectual contribution: a clean separation of *validity* of a modelling assumption from its *value*.

5.2 Why the mean dominates: a mechanism

Three forces compound. (i) *Magnitude.* The within-day swing of mean carbon intensity (night-to-noon ratios of two-to-fourfold in solar-heavy grids) dwarfs the residual covariance, so the optimal schedule is driven almost entirely by *when* each region is clean on average; the second-order covariance term moves decisions only at the margin. (ii) *Common-mode correlation.* Where correlation is strongest (US West), it is positive and shared across all zones — common, non-diversifiable risk. Diversification value requires *offsetting* fluctuations; positively co-moving regions offer no hedge, so destroying the cross-blocks (the shuffled arm) changes nothing. (iii) *Wrong tail.* Robust *tail* scheduling cares about joint *dirty* excursions, but the empirical upper tail is independent-to-decoupling (χ_U at or below Gaussian), while the co-movement that does exist is in the *clean* tail ($\chi_L > \chi_U$) — exactly the regime a risk-averse scheduler is not protecting against, and a radial asymmetry an elliptical ball cannot encode regardless. The covariance is thus both *masked* (dominated by the mean) and, for its varying part, *mis-shaped* (non-elliptical).

5.3 Relation to prior work

The finding refines rather than contradicts the carbon-aware DRO literature. Single-region robust formulations [15] remain well-motivated: our per-region marginal schedulers capture essentially all the value. The contribution is to the *multi-region* extension that treats carbon intensity as a correlated vector — a natural and previously untested generalization. We show that the covariance channel of that extension is empirically inert under realistic grid data, and we locate the reason in the mean-dominance and non-ellipticity of the dependence. This both saves practitioners from modelling complexity that does not pay and gives theorists a precise target: the value, if any, lives in non-elliptical tail structure that a Wasserstein ball over covariance cannot reach.

5.4 Practical recommendation

For carbon-aware schedulers on the grids studied, a *per-region marginal* robust scheduler — one ambiguity set per zone, no cross-region covariance — captures essentially all the achievable robust value at a fraction of the modelling and estimation cost. Estimating a full joint covariance is not merely unnecessary; in the diversification-friendly case it is mildly *counterproductive* (the joint arm fit noise and *lost* up to 0.23%). The engineering effort spared by not modelling spatial covariance is better spent on accurate per-region mean forecasts, which is where the value demonstrably lies.

5.5 Limitations

Five bound the claims. *(i) Representative schedule.* We evaluate one canonical workload (fixed daily utilization, splittable load, ramp limits); a sharply different cost geometry could in principle re-weight the covariance term, though the mean-dominance argument is scale-driven and likely to persist. *(ii) Test horizon.* The primary read is a single held-out year (2025); walk-forward to 2024 reproduces the null, but multi-year rolling evaluation would strengthen external validity. *(iii) Dependence model.* The DRO uses a covariance (second-moment) ambiguity set by construction; Phase 2 (Section 4.6) closes this gap with Gaussian and Clayton copula schedulers, but a full copula-*ambiguity* Wasserstein DRO in the Fan–Ji–Lejeune sense remains future work. *(iv) Geographic scope.* The region sets span WECC, the Eastern Interconnection, and Iberia–France across the dependence spectrum, but grids with different generation mixes (e.g. hydro-thermal systems with strong seasonal storage) remain untested. *(v) Data licence and clocks.* Carbon intensity is Electricity Maps lifecycle estimates on a common local clock per region set; alternative intensity methodologies or clock conventions could shift point estimates, though not by the order of magnitude separating the observed gaps from materiality.

6 Conclusions

6.1 Answers to the research questions

RQ1 — Is carbon intensity spatially correlated in coupled grids? Yes. After removing the shared diurnal cycle, residual cross-region correlations reach 0.78 in the US West and 0.73 in the Ontario-anchored Eastern belt and survive de-seasonalization (Section 4.1). The premise of spatial DRO is empirically sound; the engineered heterogeneous portfolio confirms the diagnostic resolves genuine near-zero dependence (0.00) when it is present.

RQ2 — Does encoding that correlation in a covariance-based ambiguity set improve robust scheduling? No. Across three primary region sets spanning the dependence spectrum (and two corroborating panels), the out-of-sample $\mathrm{CVaR}_{0.95}$ spatial gap never exceeds a few tenths of one percent and is not sign-stable; the null survives Ledoit–Wolf shrinkage, seasonal and AR(1) residualization, Benjamini–Hochberg correction over 144 cells, walk-forward validation to 2024, and a log-denser ε grid (Sections 4.2–4.3). In the most diversification-friendly case the joint covariance is mildly counterproductive. Critically, this is an *active* null: cross-validation selects a non-trivial ambiguity radius ($\varepsilon^\star = 1$) in nearly every cell of the US/Canada cases, so the DRO is genuinely robustifying and the spatial covariance still adds nothing — a stronger result than a null in which the robustification had collapsed to the nominal problem.

RQ3 — Why? Two complementary mechanisms. A mean-ablation experiment proves the covariance signal is real and worth up to 1.46% of tail emissions in a covariance-only world, but is dominated by the diurnal mean carbon field in the problem as posed (Section 4.4). A tail-dependence analysis shows the residual dependence is non-elliptical — upper-tail-independent and radially asymmetric ($\chi_L > \chi_U$) — and therefore structurally invisible to a covariance ball even where it does vary (Section 4.5). The covariance is both *masked* and *mis-shaped*. Phase 2 settles which mechanism binds: replacing the covariance ball with a Clayton copula built for the $\chi_L > \chi_U$ asymmetry still recovers no material value (Section 4.6), and Proposition 1 explains why — CVaR's translation invariance makes the dependence model second-order to the mean field. The limit is not the shape of the dependence object; it is mean-dominance.

6.2 Contributions

This thesis contributes: (1) a pre-registered falsification methodology (shuffled-marginals, extended to copula-coupled resampling) that cleanly separates the *validity* of a spatial modelling assumption from its *value*, transferable to any correlated-uncertainty DRO; (2) a replicated, robustness-checked null on whether spatial dependence improves carbon-aware DRO scheduling, holding across multiple real grids *and* across the dependence hierarchy from covariance ball to elliptical and lower-tail copulas; (3) a causal mechanism — formalized as a mean-dominance bound (Proposition 1) and demonstrated by mean-ablation — that explains the null rather than merely reporting it; and (4) a fully reproducible, version-controlled, CI-tested pipeline (168 tests) with archived license-safe summary tables for every reported number.

6.3 Future work

Phase 2 closed the most immediate question — whether a copula that sees the asymmetry recovers value (it does not) — and in doing so sharpened the remaining agenda. First, a *copula-ambiguity Wasserstein DRO* in the Fan–Ji–Lejeune sense [3], which robustifies the dependence structure itself rather than sampling from a single fitted copula; Proposition 1 predicts the mean field will still dominate, but this would make the claim fully model-free. Second, an *inter-region transfer channel*: allowing load to physically migrate between zones ($f_{r \to s,t}$) converts spatial structure from a passive hedge into an active decision lever and remains an SOCP; because the mean-dominance bound concerns *passive* risk reduction, the transfer channel is the most promising route to extracting spatial value, and it overlaps a collaborator's scope pending supervisor sign-off. Third, *rolling re-optimization* over multiple test years. The null is thus not a dead end but a sharpened mandate: it shows that the value, if any, will not come from a better passive dependence model but from an active spatial decision.

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A Reproducibility Details

Repository and environment. All code lives in a private, version-controlled repository with a continuous-integration suite (168 unit tests; the Electricity-Maps integration test is excluded from CI as it requires licensed data). Experiments run under Python with CVXPY; solvers are CLARABEL/ECOS for the second-order cone program and HiGHS for deterministic LP baselines. The licensed raw carbon-intensity data is never committed; only derived aggregate summary tables (CV-selected ε, test CVaR, spatial gaps, bootstrap CIs, BH p-values) are archived, so the non-redistribution academic licence is respected while every reported number remains traceable.

Pre-registration protocol. Each experiment follows commit-lock → dry-run → single test read. The full configuration (utilization 0.80; $\alpha \in \{0.30, 0.50, 0.75\}$; ε grid $\{0, 0.1, 1, 10, 100, 1000\}$; 5-fold blocked time-series CV; 1000 bootstrap resamples at a fixed seed; CVaR level 0.95; train 2021–2024, test 2025) is committed before any test-set evaluation, and a -dry-run gate runs cross-validation only. The 3c capacity bounds $(x_{\min}, x_{\max}) = (50, 75)$ were Goldilocks-calibrated on training data alone (bind fraction 0.28–0.37).

Reproducing the experiments. Every result regenerates with

```
python -m scripts.run_case_experiment -region-set <case> [flags]
```

where <case> is one of us_west, taskc, us_hetero, and flags select the estimator (-shrinkage, -residualize), mean-ablation (-ablate-mean), walk-forward (-test-year), or grid density (-eps-grid). Snapshot files are named <case>_regimes_<date>[_<variant>].csv, and the snapshot README maps each to its source command. Table 6 maps each results subsection to its archived table.

Table 6: Mapping of reported results to archived summary tables.

Section	Result	Archived source
4.2	Spatial gaps (Table 1)	{case}_regimes_2026-06-10.csv
4.3	Robustness (Table 2)	{case}_..._{lw,seasonal,ar1,ty2024}.csv
4.3	BH correction	bh_correction.csv
4.4	Mean-ablation (Table 3)	{case}_..._ablate-{level,flat}.csv
4.1,4.5	CA/Iberia panels	taskA_.. , es_pt_fr_..csv

Annex A: Individual Contribution Statement

This Research Capstone was completed individually by **Marco Ortiz Togashi** under the supervision of Prof. Bissan Ghaddar. As sole author, I am responsible for all components of the work:

- **Research design.** Formulated the three research questions, designed the shuffled-marginals falsification test, and specified the pre-registration protocol that separates assumption validity from value.
- **Modelling.** Specified the Mahalanobis–Wasserstein DRO scheduling model and its SOCP reformulation, the constraint regimes (including the variable carbon-coupled capacity constraint adapted from the SoCC'25 formulation), and the mean-ablation and tail-dependence diagnostics.

- **Data and implementation.** Assembled the multi-grid carbon-intensity and temperature panels, implemented the full experimental pipeline, the robustness battery, the Benjamini–Hochberg correction, and the figures, under version control with a continuous-integration test suite.
- **Analysis and writing.** Produced and interpreted all results, established the causal mechanism, and wrote the manuscript in full.

Generative AI tools were used as a coding and drafting assistant under my direction, as declared on the cover page; all research questions, modelling decisions, experimental designs, interpretations, and final text are my own. The inter-region transfer-channel proposal noted as future work overlaps a collaborator's scope and is identified as such, pending supervisor coordination.

Marco Ortiz Togashi　　　　　　Date